Analysis of fire burn severity and post-fire vegetation recovery with Landsat time-series

GEM520 Lab 8 - Final Demonstration of R skills

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Introduction

Summer 2017 was considered as one the worst wild fire season in BC (until 2023), with over 1.2 million ha of burned areas, 65,000 people displaced and $649 million spent on fire suppression (https://www2.gov.bc.ca/gov/content/safety/wildfire-status/about-bcws/wildfire-history/wildfire-season-summary).

The Prouton Lakes fire (fire ID C30870) burned part of the Alex Fraser Research Forest (AFRF) located near Williams Lake, BC, on the traditional, ancestral and unceded territory of the T'exelcemc, Xatsu'll and Esket First Nations. The burn severity varied from low-severity ground fire to high-severity canopy fire.

Lab objective

The objective of this lab is to quantify the burn severity of the Prouton Lakes fire and examine the recovery and re-establishment of vegetation in the years following the fire. You will use a Landsat 8 OLI time-series of images acquired between the 1st of July and the 31st of August between 2013 and 2021.

Description of data

Landsat time-series Landsat 8 Collection 2 Level-2 surface reflectance (SR) products are generated using the Land Surface Reflectance Code (LaSRC) algorithm that corrects for atmospheric effects at the time of the image acquisition. Unlike Level-1 products that represent top-of-atmosphere reflectance, Level-2 SR products therefore represent the fraction of incoming sunlight reflected by the Earth's surface (bottom-of-atmosphere) and can be combined and compared across time and space. Land surface temperature (ST) products are also generated in Landsat 8 Level-2 products. The surface reflectance and surface temperature products are accompanied by pixel-level quality assessment (QA) layers that indicate the presence of clouds, cloud shadows, snow/ice and water.

We focused on a single Landsat scene overlapping the study area (WRS Path 048; WRS Row 023) and downloaded all the Level-2 products acquired between the 1st of July and the 31st of August between 2013 and 2021 from the USGS Earth Explorer. This resulted in 35 products of approximately 1 GB each. Each product is named as described below.

LXSS_LLLL_PPPRRR_YYYYMMDD_yyyymmdd_CX_TX
(e.g., LC08_L2SP_039037_20150728_20200318_02_T1)

L	Landsat
X	Sensor (“O” = OLI; “T” = TIRS; “C” = OLI/TIRS)
SS	Satellite (“08” = Landsat 8)
LLLL	Processing correction level (“L2SP” if SR and ST are generated or “L2SR” if ST could not be generated)
PPP	Path
RRR	Row
YYYY	Year of acquisition
MM	Month of acquisition
DD	Day of acquisition
yyyy	Year of Level 2 processing
mm	Month of Level 2 processing
dd	Day of Level 2 processing
CX	Collection number (“02”)
TX	Collection category (“RT” = Real-Time; “T1” = Tier 1; “T2” = Tier 2)

Each product contains 19 TIF files (SR, ST and QA layers) and a Metadata file.

Band Designation	Band Name	Data Type	Units	Data Range	Valid Range	Fill Value	Multiplicative Scale Factor	Additive Offset
ProductID_SR_B1	Band 1 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B2	Band 2 SR	UINT16	Reflectance	1 - 65535	7273 – 43636	0	0.0000275	-0.2
ProductID_SR_B3	Band 3 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B4	Band 4 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B5	Band 5 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B6	Band 6 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_SR_B7	Band 7 SR	UINT16	Reflectance	1 - 65535	7273 - 43636	0	0.0000275	-0.2
ProductID_ST_B10	Band 10 ST	UINT16	Kelvin	1 - 65535	293 - 61440	0	0.00341802	149
ProductID_QA_PIXEL	Level 2 Pixel Quality Band	UINT16	Bit Index	1 - 65535	21824 – 65534	1	NA	NA
ProductID_SR_QA_AEROSOL	Aerosol QA	UINT8	Bit Index	0 - 255	1 - 255	1	NA	NA
ProductID_QA_RADSAT	Radiometric Saturation QA	UINT16	Bit Index	0 - 65535	0 - 3829	NA	NA	NA
ProductID_ST_QA	Surface Temperature Uncertainty	INT16	Kelvin	0 - 32767	0 - 32767	-9999	0.01	NA
ProductID_ST_TRAD	Thermal Radiance	INT16	W/(m^2.sr.µm)/DN	0 - 22000	0 – 22000	-9999	0.001	NA
ProductID_ST_URAD	Upwelled Radiance	INT16	W/(m^2.sr.µm)/DN	0 - 28000	0 – 28000	-9999	0.001	NA
ProductID_ST_DRAD	Downwelled Radiance	INT16	W/(m^2.sr.µm)/DN	0 - 28000	0 - 28000	-9999	0.001	NA
ProductID_ST_ATRAN	Atmospheric Transmittance	INT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMIS	Emissivity estimated from ASTER GED	INT16	Unitless	0 - 10000	0 - 10000	-9999	0.0001	NA
ProductID_ST_EMSD	Emissivity standard deviation	INT16	Unitless	0 - 32767	0 – 10000	-9999	0.0001	NA
ProductID_ST_CDIST	Pixel distance to cloud	INT16	Kilometers	0 - 24000	0 – 24000	-9999	0.01	NA
ProductID_MTL	Level 2 Metadata file	NA	NA	NA	NA	NA	NA	NA

Table 6-1. Collection 2 (C2) Landsat 8 Level 2 Science Products (L2SP) Band Specifications

The surface reflectance measured in the spectral bands of the OLI sensor are labeled as SR_B# where # is the band number

Band Number	Band Name	Wavelength (um)	Spatial resolution (m)
1	Band 1 - Coastal aerosol	0.43-0.45	30
2	Band 2 - Blue	0.45-0.51	30
3	Band 3 - Green	0.53-0.59	30
4	Band 4 - Red	0.64-0.67	30
5	Band 5 - Near Infrared (NIR)	0.85-0.88	30
6	Band 6 - SWIR 1	1.57-1.65	30
7	Band 7 - SWIR 2	2.11-2.29	30

The `QA_PIXEL` layer indicates the presence of clouds, cirrus, snow/ice and water as well as confidence levels in the presence of each. A 16-digit binary number is used to encode each condition.

Bit	Flag Description	Values
0	Fill	0 for image data 1 for fill data
1	Dilated Cloud	0 for cloud is not dilated or no 1 for cloud dilation
2	Cirrus	0 for Cirrus Confidence: no co set or Low Confidence 1 for high confidence cirrus
3	Cloud	0 for cloud confidence is not h 1 for high confidence cloud
4	Cloud Shadow	0 for Cloud Shadow Confiden 1 for high confidence cloud sh
5	Snow	0 for Snow/Ice Confidence is r 1 for high confidence snow co
6	Clear	0 if Cloud or Dilated Cloud bits 1 if Cloud and Dilated Cloud b
7	Water	0 for land or cloud 1 for water
8-9	Cloud Confidence	00 for no confidence level set 01 Low confidence 10 Medium confidence 11 High confidence
10-11	Cloud Shadow Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence
12-13	Snow/Ice Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence
14-15	Cirrus Confidence	00 for no confidence level set 01 Low confidence 10 Reserved 11 High confidence

Table 6-2. Landsat 8 Pixel Quality Assessment (QA_PIXEL)

We show below how pixel values are determined for 4 conditions: clear, water, high confidence cloud and high confidence cloud shadow. Additional examples are provided in the Table 6-3 of the Landsat 8 Collection 2 Level-2 Science Product Guide

15	14	13	12	11	10	9	8	7	6	5	
Cirrus Conf.		Snow/Ice Conf.		Cloud Shadow Conf.		Cloud Conf.		Water	Clear	Snow	C Sh

Clear conditions

0	1	0	1	0	1	0	1	0	1	0	

Binary value = 0101010101000000
Pixel value = 21824

Water

0	1	0	1	0	1	0	1	1	0	0	

Binary value = 0101010110000000
Pixel value = 21888

High Confidence Cloud

0	1	0	1	0	1	1	1	0	0	0	

Binary value = 0101011100001000
Pixel value = 22280

High Confidence Cloud Shadow

0	1	0	1	1	1	0	1	0	1	0	

Binary value = 0101110101010000
Pixel value = 23888

Historical fire perimeters A polygon shapefile showing the perimeter of historical wildfires in BC (all wildfires before current year) is available on the BC Data Catalogue. This layer is available in `"data/PROT_HISTORICAL_FIRE_POLYS_SP/H_FIRE_PLY_polygon.shp"`. We describe a few fields below:

- `FIRE_NO`: A unique ID assigned to each layer (e.g. C30870 for the Prouton Lakes fire)
- `FIRE_YEAR`: The year when the fire happened
- `FIRE_CAUSE`: The cause of the fire (either `Lightning`, `Person` or `Unknown`)
- `SIZE_HA`: The area of each fire, in hectares (ha)

Instructions

Step 1 - Historical fires in BC **Provide code AND written answers for the following questions**. Use the `H_FIRE_PLY_polygon.shp` shapefile for this step.

Attaching packages

You can attach the package that you are using in the following code block

```r
library(terra)    # rasters
```

```
## Warning: package 'terra' was built under R version 4.5.1
```

```r
library(sf)       # vectors
```

```
## Warning: package 'sf' was built under R version 4.5.1
```

```r
library(dplyr)    # data wrangling
```

```
## Warning: package 'dplyr' was built under R version 4.5.1
```

```r
library(ggplot2)  # plotting
```

```
## Warning: package 'ggplot2' was built under R version 4.5.1
```

```r
library(stringr)  # string helpers
```

```
## Warning: package 'stringr' was built under R version 4.5.1
```

```r
library(tidyr)    # extra wrangling
```

```
## Warning: package 'tidyr' was built under R version 4.5.1
```

```r
library(lubridate)
```

```
## Warning: package 'lubridate' was built under R version 4.5.1
```

- What was the number of fires and total land area burned (in hectares) in 2017?

```r
bc_fires <- st_read("data/PROT_HISTORICAL_FIRE_POLYS_SP/H_FIRE_PLY_polygon.shp", quiet = TRUE)
bc_fires_2017 <- bc_fires %>%
  filter(FIRE_YEAR == 2017)
fires_2017 <- nrow(bc_fires_2017)
fires_2017
```

```
## [1] 342
```

```r
# Total land area burned in 2017
burned_2017 <- sum(bc_fires_2017$SIZE_HA, na.rm = TRUE)
burned_2017
```

```
## [1] 1222205
```

- What is the ID and burned area (in hectares) of the three largest fires recorded in 2017?

```r
largest_2017 <- bc_fires_2017 %>%
  arrange(desc(SIZE_HA)) %>%
  slice(1:3) %>%
  select(FIRE_NO, SIZE_HA)

largest_2017
```

```
## Simple feature collection with 3 features and 2 fields
## Geometry type: MULTIPOLYGON
## Dimension:     XY
## Bounding box:  xmin: 1100547 ymin: 642677.6 xmax: 1355614 ymax: 905150.6
## Projected CRS: NAD83 / BC Albers
##   FIRE_NO  SIZE_HA                       geometry
## 1  C10784 520885.2 MULTIPOLYGON (((1172964 875...
## 2  C50647 239339.6 MULTIPOLYGON (((1220745 798...
## 3  K20637 192016.5 MULTIPOLYGON (((1349947 733...
```

- How much land area was burned by the Prouton Lakes fire (ID C30870) ?

```r
prouton_fire <- bc_fires %>%
  filter(FIRE_NO == "C30870") %>%
  pull(SIZE_HA)

prouton_fire
```

```
## [1] 859.3
```

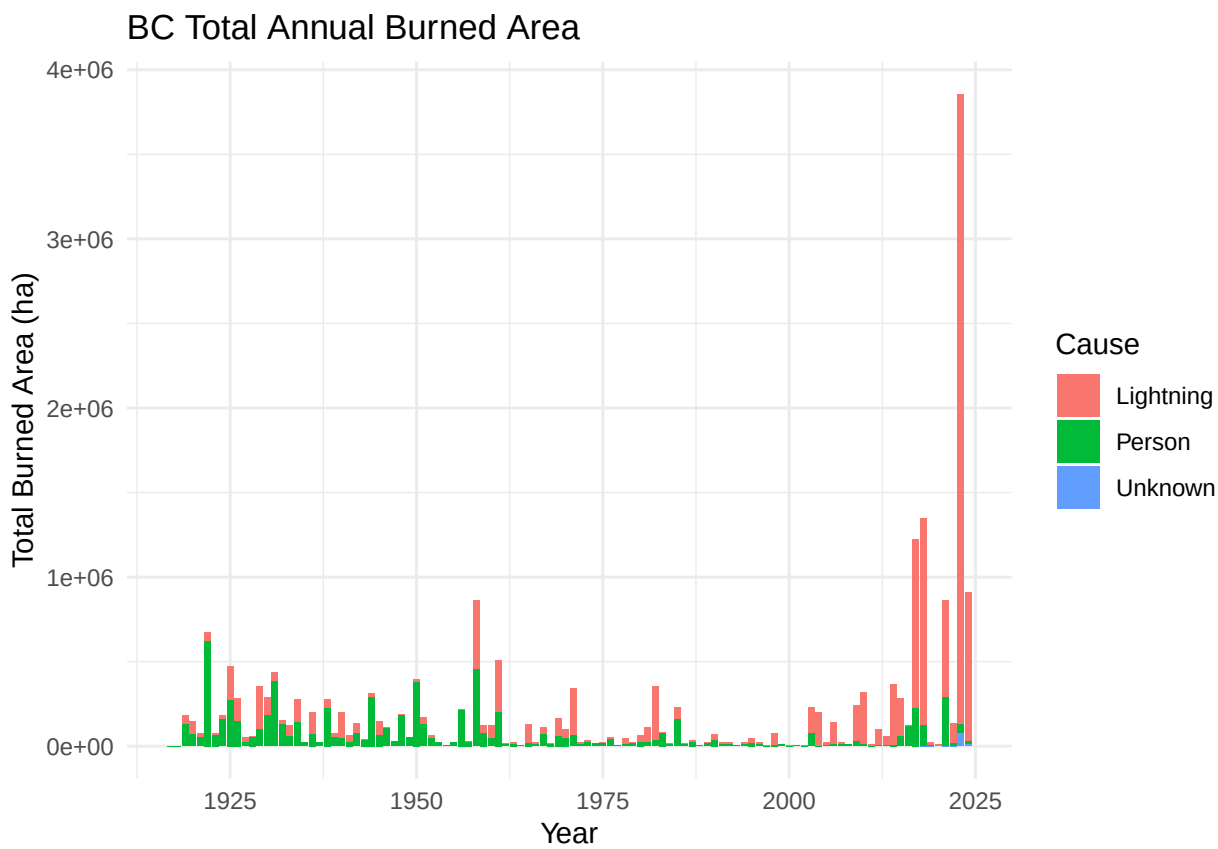
Include the following figure

Use `ggplot2` to create a barplot (examples here) showing the total area burned per year in BC. Each bar should be colored by fire cause.

Hints

- Summarizing a `sf` objects is faster if the `geometry` information is *dropped*

```r
total_annual <- bc_fires %>%
  group_by(FIRE_YEAR, FIRE_CAUSE) %>%
  summarise(total_area = sum(SIZE_HA, na.rm = TRUE), .groups = "drop")

ggplot(total_annual, aes(x = FIRE_YEAR, y = total_area, fill = FIRE_CAUSE)) +
geom_bar(stat = "identity") +
labs(title = "BC Total Annual Burned Area",
x = "Year", y = "Total Burned Area (ha)", fill = "Cause") +
theme_minimal()
```



Step 2 - Pre-processing Level-2 products and calculating vegetation indices For each L2 product, put each surface reflectance band (Band 1 to Band 7) in the same raster object, **mask** pixels that do not have the value `21824` (*clear conditions*) in the `QA_PIXEL` layers and **crop** (no need to mask) the multi-layer raster to the Prouton Lakes fire extent. You also need to calculate the normalized vegetation difference index (NDVI) and the normalized burn ratio (NBR) for each cropped and masked raster (see hints). For each product, save the multi-layer SR product, NDVI and NBR layers in the `outputs` directory with the file name `ProductID_SR.tif`, `ProductID_NDVI.tif` or `ProductID_NBR.tif` (e.g. `LC08_L2SP_048023_20130701_20200912_02_T1_NBR.tif`)

Include the following figure

A true color composite image of the Prouton Lakes fire area before (**July 7 2015**) and after the fire (**July 15 2018**)

Hints

- The functions `list.files(..., pattern = ..., full.names = TRUE)` and `list.dirs(..., full.names = TRUE, recursive = FALSE)` can be used to list all the files and directories in a path.
- The function `basename(x)` can be used to keep only a file or directory name without the preceding path

```
basename("path/to/dir/filename.tif")
```

```
## [1] "filename.tif"
```

```
basename("path/to/dir")
```

```
## [1] "dir"
```

- The function `tools::file_path_sans_ext()` can be used to remove the extension (e.g. `.TIF`) from a file name

```
tools::file_path_sans_ext("filename.tif")
```

```
## [1] "filename"
```

- The function `file.path()` can be used to construct a file path from characters.

```
file.path("path", "to", "directory", "filename.tif")
```

```
## [1] "path/to/directory/filename.tif"
```

```
file.path("path/to/directory", "filename.tif")
```

```
## [1] "path/to/directory/filename.tif"
```

- The function `paste0()` can be used to concatenate characters

```
paste0("filename", ".tif")
```

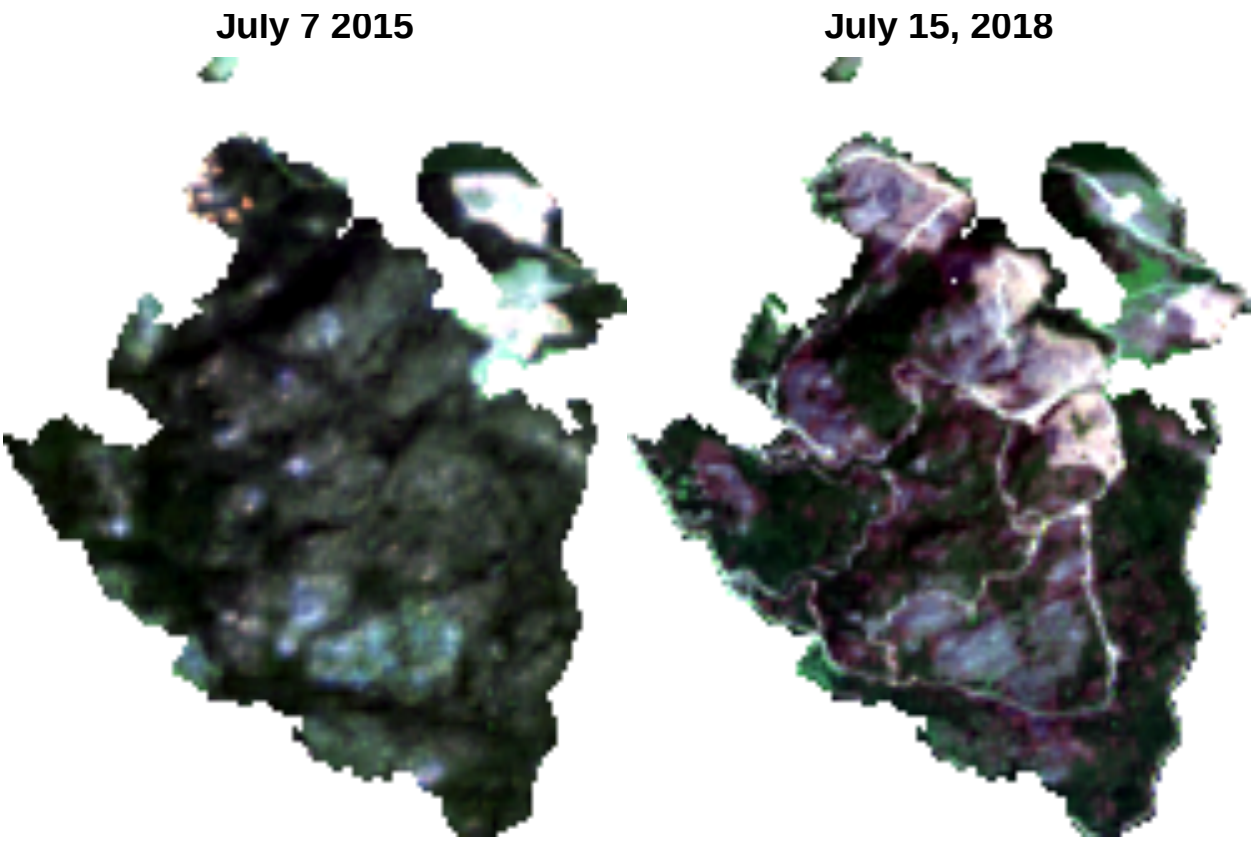
```
## [1] "filename.tif"
```

- The function `terra::mask()` has an argument named `inverse`
- The function `terra::writeRaster(x, filename = fname, overwrite = TRUE)` is used to write a `SpatRaster` to a file on disk named `fname` (containing both the path and the file name with extension). The argument `overwrite = TRUE` allows `writeRaster` to overwrite an existing file with the same name.
- $$NDVI = \frac{NIR - RED}{NIR + RED}$$
- $$NBR = \frac{NIR - SWIR2}{NIR + SWIR2}$$

```r
landsat <- "data/Landsat 8 OLI_TIRS C2 L2"
layers <- list.dirs(landsat, recursive = FALSE)
fires_bound <- vect(bc_fires)
prouton <- fires_bound[fires_bound$FIRE_NO == "C30870", ]
```

```r
output <- "output/"

for (lyr in layers) {
  layer_id <- basename(lyr)
  band_files <- list.files(lyr, pattern = "_SR_B[1-7].TIF$", full.names = TRUE)
  bands <- rast(band_files)

  qa_file <- list.files(lyr, pattern = "_QA_PIXEL.TIF$", full.names = TRUE)
  qa <- rast(qa_file)
  mask <- qa == 21824
  bands_masked <- mask(bands, mask, maskvalues = FALSE)


  prouton_proj <- project(prouton, bands)
  bands_cropped <- mask(bands_masked, prouton_proj)
  nir <- bands_cropped[[5]]
  red <- bands_cropped[[4]]
  swir2 <- bands_cropped[[7]]
  ndvi <- (nir - red) / (nir + red)
  nbr <- (nir - swir2) / (nir + swir2)
  sr_out <- file.path(output,"LC08_L2SP_048023_SR", paste0(layer_id,
  "_SR.tif"))
  ndvi_out <- file.path(output,"LC08_L2SP_048023_NDVI", paste0(layer_id,
  "_NDVI.tif"))
  nbr_out <- file.path(output,"LC08_L2SP_048023_NBR", paste0(layer_id,
  "_NBR.tif"))

  writeRaster(bands_cropped, sr_out, overwrite = TRUE)
  writeRaster(ndvi, ndvi_out, overwrite = TRUE)
  writeRaster(nbr, nbr_out, overwrite = TRUE)
}

sr_files <- list.files(file.path(output, "LC08_L2SP_048023_SR"),
                       pattern = "_SR.tif$",
                       full.names = TRUE)
scene_dates <- substr(basename(sr_files), 18, 25)
before_scene <- rast(sr_files[scene_dates == "20150707"])
after_scene <- rast(sr_files[scene_dates == "20180715"])
par(mfrow = c(1, 2))

plotRGB(before_scene,
        r = 4, g = 3, b = 2,
        stretch = "lin",
        main = "July 7 2015",
        ext = prouton_proj)

plotRGB(after_scene,
        r = 4, g = 3, b = 2,
```

```
        stretch = "lin",
        main = "July 15, 2018",
        ext = prouton_proj)
```



Step 3 - Yearly composites of NDVI and NBR For each year in the NDVI and NBR time-series, create yearly composites by calculating the average value of each pixel from all the layers acquired in July and August. Make sure that `NA` values are not included in the average.

Then, calculate the average NDVI of all the pixels overlaid by the Prouton Lakes fire, for each year in the time series.

Include the following figure

Use `ggplot2` to create a connected scatterplot (points joined by lines) showing the average NDVI across the Prouton Lakes fire area through time. tc_2015 <- rast("outputs/LC08_...20150707..._SR.tif") tc_2018 <- rast("outputs/LC08_...20180715..._SR.tif")

RGB 4-3-2

plotRGB(tc_2015, r = "SR_B4", g = "SR_B3", b = "SR_B2", stretch = "lin") plotRGB(tc_2018, r = "SR_B4", g = "SR_B3", b = "SR_B2", stretch = "lin")

Hint

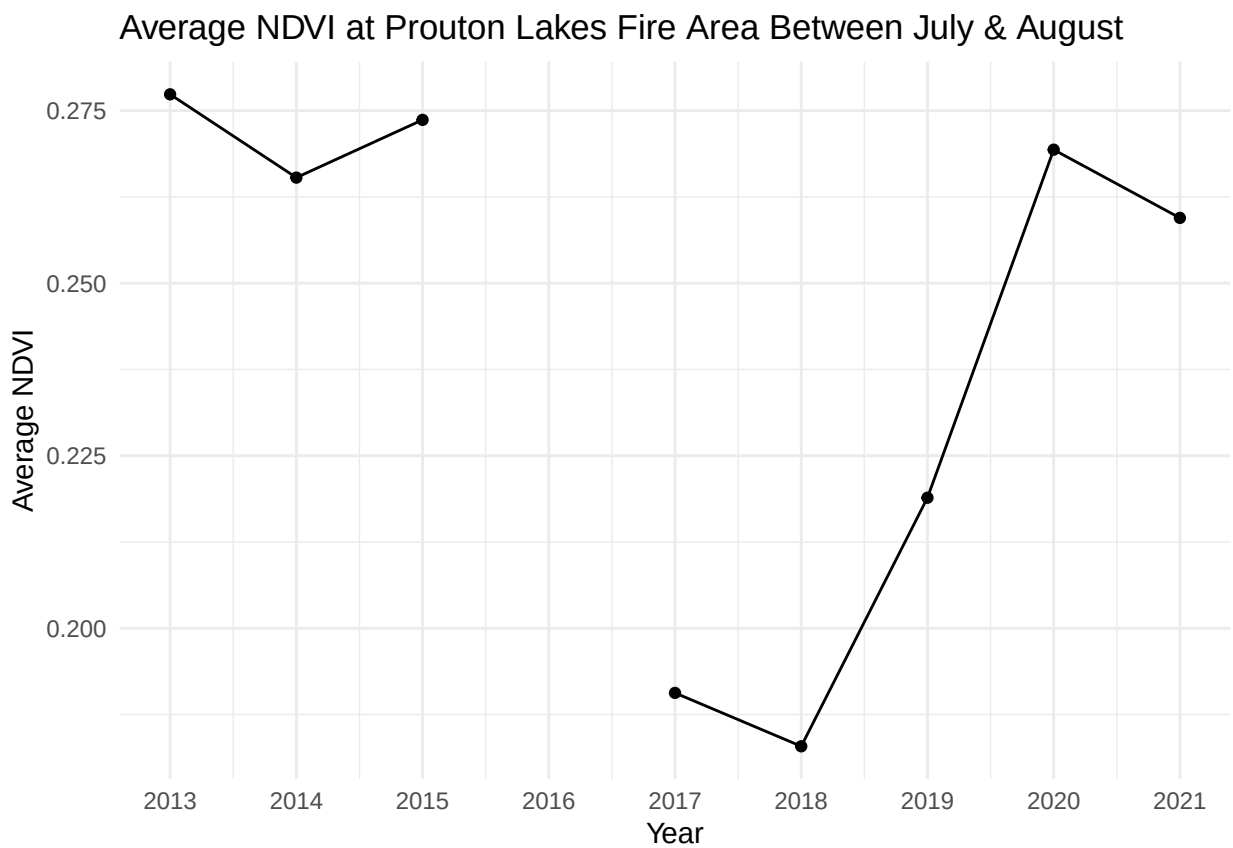
`geom_line()` requires the `group` aesthetic to be set : `aes(x = ..., y = ..., group = ...)`. If there is no variable to group the lines by (i.e. a single group of data connected by lines), you can use `group = 1` (`aes(x = ..., y = ..., group = 1)`)

```
ndvi <- "output/LC08_L2SP_048023_NDVI"
nbr <- "output/LC08_L2SP_048023_NBR"
ndvi_lyrs <- list.files(ndvi, "_NDVI.tif$", full.names = TRUE)
nbr_lyrs <- list.files(nbr, "_NBR.tif$", full.names = TRUE)
scene_dates <- substr(basename(ndvi_lyrs), 18, 25)
dates <- as.Date(scene_dates, format = "%Y%m%d")
years <- unique(year(dates))
ndvi_stats <- data.frame(year = integer(), mean_ndvi = numeric())

for (year in years) {
  ndvi_stack <- rast(ndvi_lyrs[year(dates) == year])
  ndvi_comp <- mean(ndvi_stack, na.rm = TRUE)
  ndvi_comp <- crop(ndvi_comp, prouton_proj)
  val <- terra::extract(ndvi_comp,
                        prouton_proj,
                        fun = mean, na.rm = TRUE)[,2]

  ndvi_stats <- data.frame(
    year = c(ndvi_stats$year, year),
    mean_ndvi = c(ndvi_stats$mean_ndvi, val)
)

  nbr_stack <- rast(nbr_lyrs[year(dates) == year])
  nbr_stack_idx <- year(dates) == year
  nbr_comp <- mean(nbr_stack, na.rm = TRUE)
  nbr_comp
}


ndvi_stats$year
```

```
## [1] 2013 2014 2015 2016 2017 2018 2019 2020 2021
```

```
ggplot(ndvi_stats, aes(x = year, y = mean_ndvi)) +
geom_point() +
geom_path() +
scale_x_continuous(breaks = ndvi_stats$year) +
labs(title = "Average NDVI at Prouton Lakes Fire Area Between July & August",
x = "Year", y = "Average NDVI") +
theme_minimal()
```

```
## Warning: Removed 1 row containing missing values or values outside the scale range
## (`geom_point()`).
```



Step 4 - Classify burn severity The burn severity of a fire can be estimated with the delta NBR (dNBR), which is the difference between pre-fire and post-fire NBR. Generally speaking, the higher dNBR is, the more severe the fire burned the land.

$$dNBR = NBR_{prefire} - NBR_{postfire}$$

Field-based surveys are generally necessary to assess the burn severity of a fire. *Supervised classification* can then be used to create maps of burn severity. In our case, we will classify burn severity of the Prouton Lakes fire using dNBR thresholds. The reclassification table is described below:

dNBR range	Burn Severity
-0.2 to 0.15	unburned
0.15 to 0.25	low severity
0.25 to 0.3	medium severity
0.3 to 1	high severity

Considering `2015` as the pre-fire year and `2018` as the post-fire year, create a raster of the Prouton Lakes fire burn severity. Make sure that the value of each pixel is the burn severity class (i.e. `"unburned"`, `"low severity"`, `"medium severity"`, `"high severity"`) instead of an arbitrary number (see hint).

Include the following figure

A plot of the dNBR raster classified into burn severity classes, overlaid with the boundary of the Prouton Lakes fire. The legend should show the burn severity levels (i.e. `"unburned"`, `"low severity"`, `"medium severity"`, `"high severity"`)

Hints

When a `SpatRaster` `x` has discrete values that represent classes (e.g. binary values representing water or land), you can set the `levels` of `x` as follows:

```
levels(x) <- data.frame(id = c(0, 1),
                        label = c("water",
                                  "land"))
```

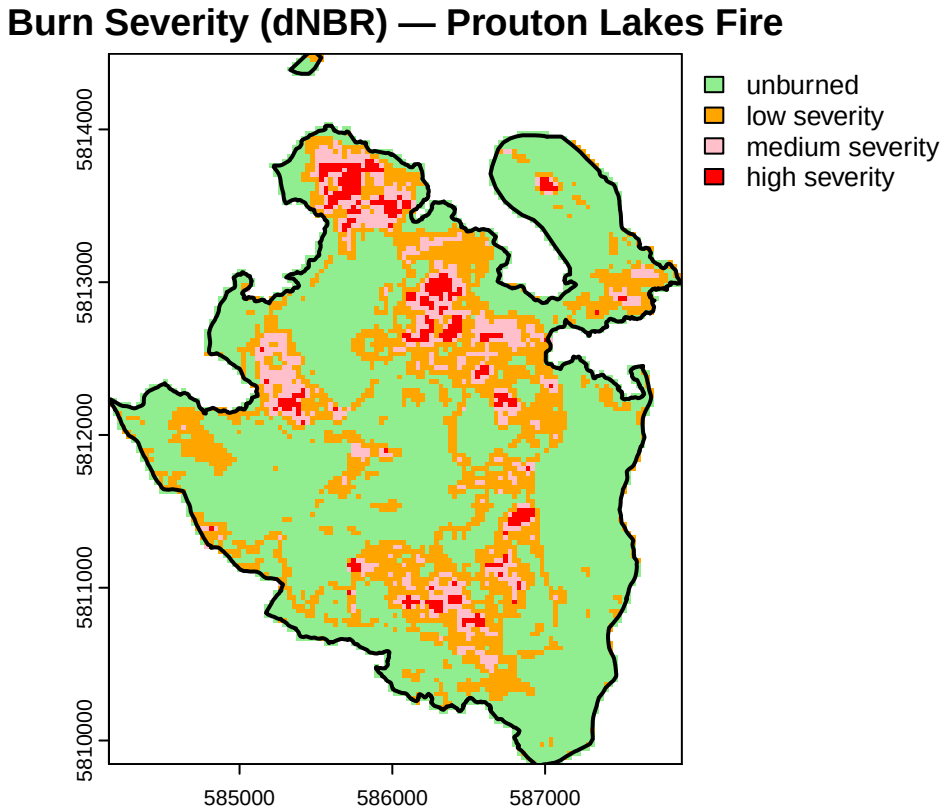
That way, values are transformed to a `factor`.

```
nbr_scenes <- substr(basename(nbr_lyrs), 18, 25)
nbr_dates <- as.Date(nbr_scenes, format = "%Y%m%d")
unique(year(nbr_dates))
```

```
## [1] 2013 2014 2015 2016 2017 2018 2019 2020 2021
```

```
nbr_2015_rast <- rast(nbr_lyrs[year(nbr_dates) == 2015])
nbr_2018_rast <- rast(nbr_lyrs[year(nbr_dates) == 2018])
nbr_2015 <- mean(nbr_2015_rast, na.rm = TRUE)
nbr_2018 <- mean(nbr_2018_rast, na.rm = TRUE)
```

```
nbr_2015 <- crop(nbr_2015, prouton_proj)
nbr_2018 <- crop(nbr_2018,prouton_proj)

dnbr <- nbr_2015 - nbr_2018
dnbr <-mask(dnbr,prouton_proj)

# reclassify
rcl <- matrix(c(-Inf, 0.15, 1,
                0.15, 0.25, 2,
                0.25, 0.3, 3,
                0.3, Inf, 4),
              ncol = 3, byrow = TRUE)

dnbr_class <- classify(dnbr, rcl)
levels(dnbr_class) <- data.frame(id = 1:4,
                                 label = c("unburned",
                                           "low severity",
                                           "medium severity",
                                           "high severity"))
plot(dnbr_class,
     col = c("lightgreen", "orange", "pink", "red"),
     main = "Burn Severity (dNBR) - Prouton Lakes Fire")
lines(prouton_proj, col = "black", lwd = 2)
```



Step 5 - Post-fire vegetation recovery Convert the burn severity raster to polygons (one polygon per burn severity class) and extract the NDVI values in 2018, 2019, 2020 and 2021 for each burn severity class. **Do not summarize the pixel values (e.g. mean). We want the value of every pixel in each burn severity class because we want to examine the distribution of NDVI values per burn severity class.**

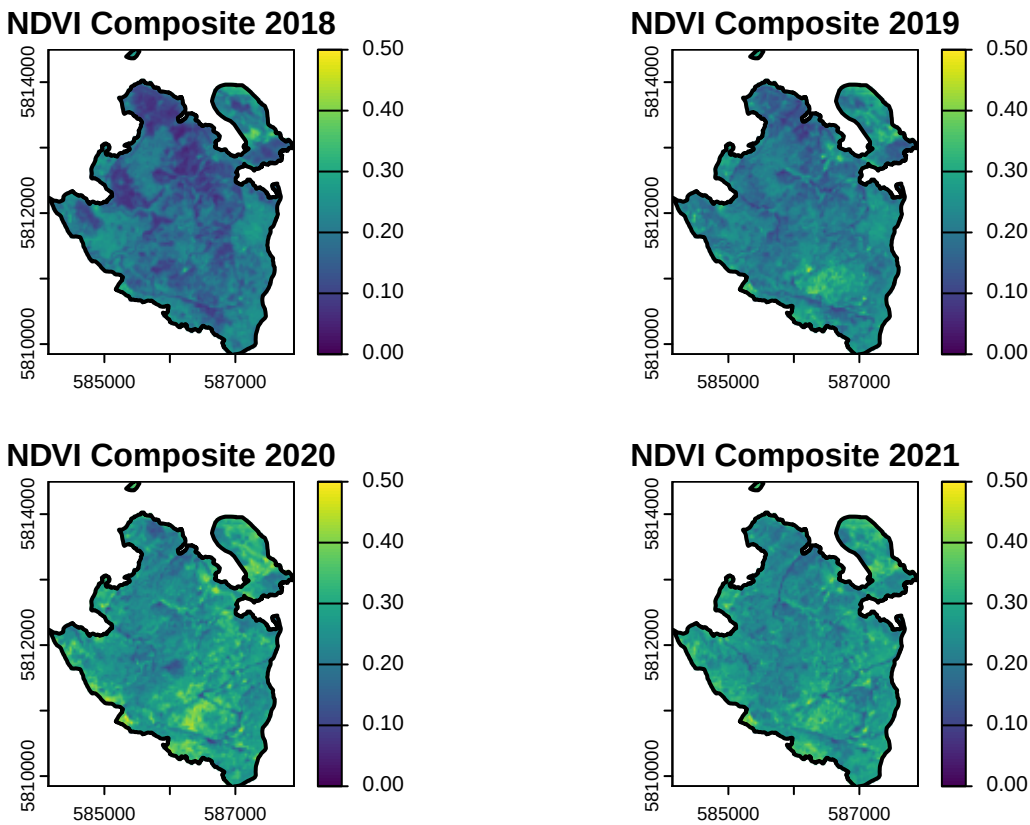
Hints

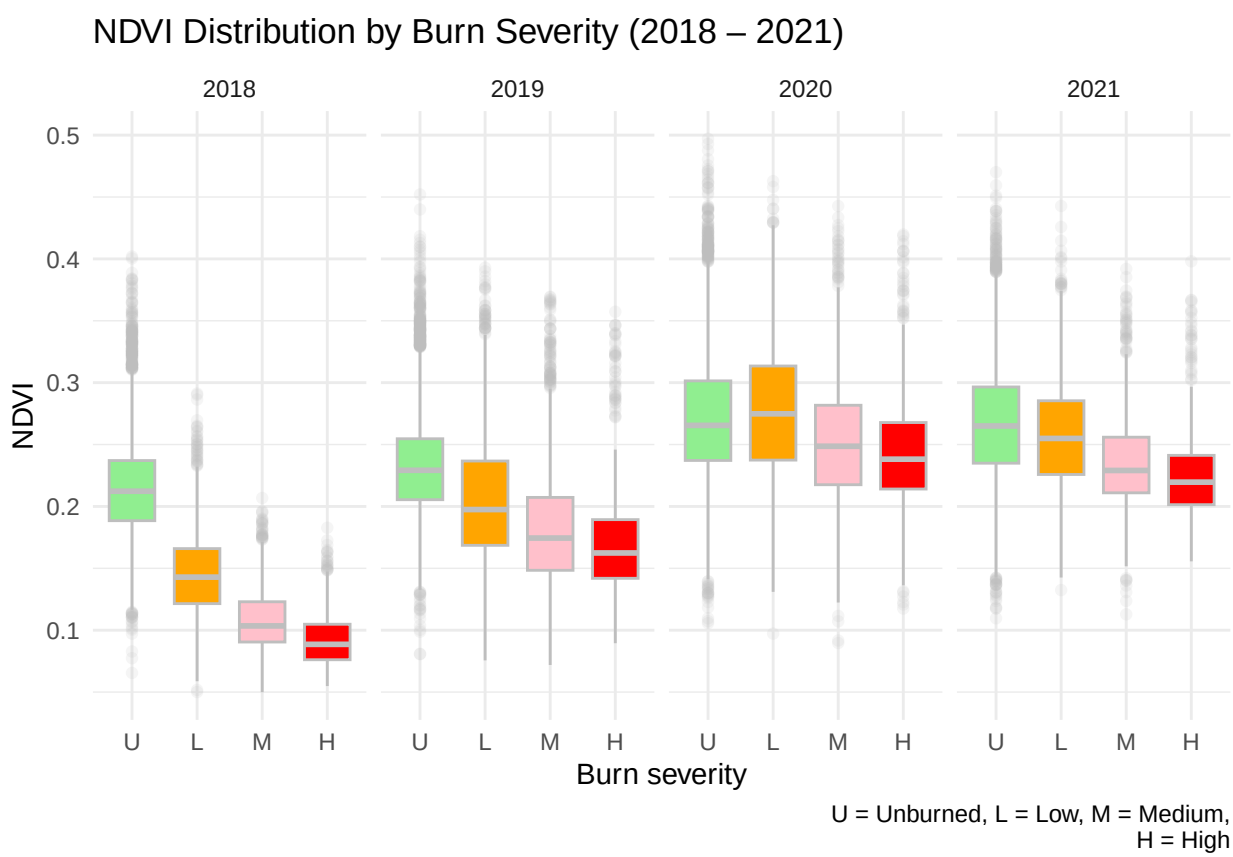
- The function `terra::as.polygons` can be used to convert a `SpatRaster` to a polygon `SpatVector`. With the default argument `dissolve = TRUE`, the cells of the `SpatRaster` with the same values will be combined to a `SpatVector`

Include the following figures

- A plot showing the yearly NDVI composites in 2018, 2019, 2020 and 2021. Make sure that the legends of the 4 plots display the same range of values between 0 and 0.5 (`plot(..., range = c(0, 0.5))`)
- Use `ggplot2` to create a boxplot (examples here) showing the distribution of NDVI values per burn severity classes in 2018, 2019, 2020 and 2021. You can either make facets for each year (`facet_wrap()`) or make one plot with different colors for each boxplot. Make sure that the burn severity classes on the x axis area ordered from unburned to high severity (**hint: make sure that burn severity is an ordered factor**)

```r
#### STEP 5 - POST-FIRE VEGETATION RECOVERY ----

target_yr <- c(2018, 2019, 2020, 2021)
levels <- c("unburned", "low severity", "medium severity", "high severity")
ndvi_composites <- list()

for (year in target_yr) {
  year_mask <- lubridate::year(dates) == year
  files_yr <- ndvi_lyrs[year_mask]
  stopifnot(length(files_yr) > 0)
  ndvi_stack <- rast(files_yr)
  ndvi_comp <- mean(ndvi_stack, na.rm = TRUE)
  names(ndvi_comp) <- paste0("ndvi_", year)
  ndvi_comp <- crop(ndvi_comp, prouton_proj)
  ndvi_composites[[as.character(year)]] <- ndvi_comp
}

burn_polys <- as.polygons(dnbr_class, dissolve = TRUE)
names(burn_polys) <- "class_id"
burn_polys <- burn_polys[!is.na(burn_polys$class_id), ]
burn_polys$severity <- factor(burn_polys$class_id,
                              levels = levels,
                              ordered = TRUE)

burn_polys$poly_id <- seq_len(nrow(burn_polys))
par(mfrow = c(2, 2))

for (year in target_yr) {
  ndvi_yr <- ndvi_composites[[as.character(year)]]
  plot(ndvi_yr, range = c(0, 0.5), main = paste("NDVI Composite", year))
  if (inherits(prouton_proj, "SpatVector")) {
  lines(prouton_proj, col = "black", lwd = 2)
  }
}
```




```
par(mfrow = c(1, 1))
ndvi_by_class <- data.frame()
for (year in target_yr) {
  ndvi_yr <- ndvi_composites[[as.character(year)]]
  burn_polys_proj <- project(burn_polys, ndvi_yr)
  stopifnot(all(c("poly_id", "severity") %in% names(burn_polys_proj)))
  ext <- terra::extract(ndvi_yr, burn_polys_proj, df = TRUE)
  val_col <- setdiff(names(ext), "ID")[1]
  ext$ndvi <- ext[[val_col]]
  ext <- ext[is.finite(ext$ndvi), ]
  attrs <- as.data.frame(burn_polys_proj)[, c("poly_id", "severity")]
  attrs$ID <- seq_len(nrow(burn_polys_proj))
  ext <- ext %>%
  left_join(attrs[, c("ID", "poly_id", "severity")], by = "ID") %>%
  mutate(year = year)

  ndvi_by_class <- bind_rows(ndvi_by_class,
                             ext[, c("poly_id",
                                     "ndvi",
                                     "severity",
                                     "year")])
}

ndvi_by_class <- ndvi_by_class[!is.na(ndvi_by_class$severity), ]

ndvi_by_class$severity <- factor(ndvi_by_class$severity,
```

```r
                                  levels = c("unburned", "low severity",
                                             "medium severity", "high severity"),
                                  labels = c("U", "L", "M", "H"),
                                  ordered = TRUE)

ggplot(ndvi_by_class, aes(x = severity, y = ndvi, fill = severity)) +
  geom_boxplot(outlier.alpha = 0.15, width = 0.7, color = "gray") +
  facet_wrap(~year, nrow = 1) +
  scale_fill_manual(values = c("U" = "lightgreen",
                               "L" = "orange",
                               "M" = "pink",
                               "H" = "red")) +
  labs(
    title = "NDVI Distribution by Burn Severity (2018 - 2021)",
    x = "Burn severity", y = "NDVI",
    caption = "U = Unburned, L = Low, M = Medium,
    H = High"
  ) +
  theme_minimal() +
  theme(
  legend.position = "none",
  axis.text.x = element_text(size = 9)
)
```